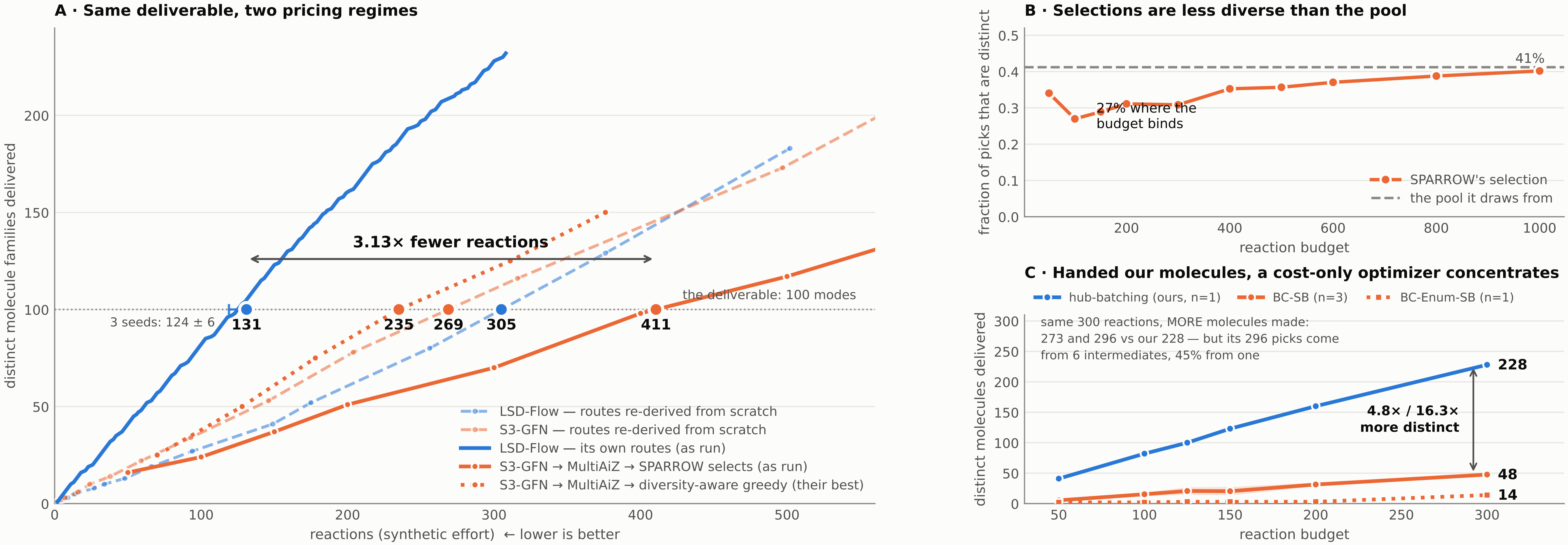
Priced the way each pipeline is actually run, LSD-Flow delivers the same 100-family library for 3.1× fewer reactions — the opposite of what the earlier from-scratch pricing showed, because that regime measured whose chemistry the planner recognised



A/B — SCENT sEH vs S3-GFN (500 candidates, 478 routed) · reward > 7.0 · τ = 0.5 · seed 42. Both arms priced by SPARROW; our readout is the independent SPARROW audit of the same library (our own count-once estimate reads 125, agreeing to 4.8%). Our side also has 3 seeds (124 ± 6, error bar) while the competitor is n=1; the quoted 131 is the WORST of our three seeds, so the ratio drawn is the conservative one. Conservative reading— our 131 vs the baseline's STRONGEST tested configuration (235: MultiAiZ + a diversity-aware selection) = 1.79×.

C — the same optimizer on OUR molecules: BC-SB over 21,001 distinct candidates (n=3 seeds); BC-Enum-SB over 131,474 molecules enumerated from the 64 intermediates a naive best-score walk collects (n=1). Raw counts, not per-distinct ratios: at tight budgets BC-Enum-SB delivers 2–3 distinct molecules, so a ratio there swings ~2× on one molecule. Ours in C is the seed-42 curve, count-once priced (the SPARROW audit of the same libraries agrees to 0.8-4.8% across our three seeds).

WHERE WE LOSE, plainly — per molecule MADE the optimizer is cheaper: 1.10 (BC-SB) and 1.01 (BC-Enum-SB) reactions/molecule versus our 1.32. It is explicitly minimizing that and has no reason to care about anything else; the claim is cost per DISTINCT molecule. Other asymmetries reported, not matched — pool 500 vs 26,069; surrogate calls ~500 vs 155,764; route planning 2.25 h vs 0.